

KHOI NGUYEN

Electrical Engineering Student — Digital ASIC, VLSI & Physical Design

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Experience

University of Waterloo

April 2026–Sept 2026

ASIC Design Co-op / Visiting Research Fellow – CMOS Design and Reliability Group

Waterloo, ON

- Owned backend physical design for a multi-block RISC-V SoC in GlobalFoundries 12LP+, driving synthesis, floorplanning, and timing closure in Cadence Genus/Innovus across concurrent design blocks.
- Completed full RTL-to-GDSII implementation and tapeout across the Cadence digital flow (Genus, Innovus, Tempus, Voltus, Pegasus, Modus), achieving signoff closure for chip submission to fabrication.
- Built and validated production-grade DFT infrastructure in Cadence Modus – scan insertion, ATPG pattern generation, and fault-simulation debug – meeting test coverage sign-off criteria used for tapeout readiness.
- Designed and verified standard-cell layouts (DFFs, SRAM bitcells) in 12nm FinFET using Cadence Virtuoso, cutting cell area by 10% while holding DRC/LVS clean.

Nano-Systems Fabrication Laboratory (Class 10 Cleanroom)

May 2025–April 2026

Student Researcher

University of Manitoba

- Executed end-to-end CMOS and MEMS fabrication in a Class 10 cleanroom, including 200 nm DUV photolithography, ion implantation, RIE/ICP etching, and PVD/CVD thin-film deposition.
- Developed and optimized 200 nm DUV lithography processes (alignment, exposure dose, focus tuning), achieving high pattern fidelity and sub-micron critical dimension (CD) control.
- Optimized plasma etch and lithography recipes, reducing defect rate by 30–50% and improving etch throughput by 40%.

Projects

NVIDIA NVDLA AI Accelerator: Full-Chip Implementation in 12-nm FinFET | *Cadence Genus, Modus, Innovus*

- Drove full-chip implementation of the open-source NVIDIA NVDLA deep-learning accelerator from RTL through GDSII in 12-nm FinFET, synthesized with Cadence Genus using a hierarchical per-partition flow, closing timing at 1.1 GHz across 2 clock domains at 7.8W.
- Generated custom SRAM macros using a foundry-provided 2-port memory compiler (RVT/MVT), replacing NVDLA's behavioral RAM models with synthesizable hard-macro views to unblock downstream synthesis and place-and-route.
- Achieved 1.11 GHz timing closure (0 ps slack) across all five partitions at the worst-case LVT corner with 900K–1.9M cell instances per partition.

Adaptive DVFS RISC-V CPU with Timing-Error Detection in 12nm FinFET | *Verilog, Cadence Genus/Innovus/Modus*

- Architected timing-error detection flip-flops to enable adaptive dynamic voltage-frequency scaling (DVFS), targeting energy efficiency gains under variable workload.
- Engineered custom scan chain/DFF architectures across 6 fault-detection schemes and built full DFT infrastructure in Cadence Modus, achieving 90%+ combined static and dynamic fault coverage.
- Closed timing at 1 GHz with zero violations across 4,739 paths via RTL synthesis in Cadence Genus using MMMC constraints and full LEF/technology library integration.
- Executed full-chip implementation in Cadence Innovus, including floorplanning, power planning, placement, CTS.
- Delivered a 9,070.45 μm^2 implementation with 15,319 leaf instances, from RTL to GDSII, in GlobalFoundries 12LP+.

CMOS Device Fabrication & Process Integration using 200 nm DUV | *Nano Fabrication*

- Fabricated CMOS transistors using 200 nm DUV lithography on 6-inch wafers with overlay alignment and integrated metal interconnect patterning.
- Performed dry etch process (SF₆, CF₄, O₂) on ICP-RIE systems to achieve precise sub-micron feature definition.

Technical Skills

Digital ASIC and Physical Design: Verilog, RISC-V architecture, Cadence Genus, Innovus, synthesis, floorplanning, place and route (PnR), clock tree synthesis (CTS), static timing analysis (STA), Design for Test (DFT), ATPG

Analog & Mixed-Signal IC Design: Cadence Virtuoso, ADE/Spectre, nanoscale CMOS design (12 & 14 nm FinFET, 12 nm FDSOI, 45 nm, 65 nm), full-custom layout techniques, parasitic extraction, DRC/LVS verification

Semiconductor Fabrication: CMOS & MEMS process integration, photolithography, ion implantation, thin-film deposition (CVD/PVD), RIE, ICP etching, SEM, profilometry, ellipsometry

Programming and Systems: Python, TCL, MATLAB, Linux/Unix, ARM assembly

Education

University of Manitoba

Aug 2023–Dec 2027

Bachelor of Science in Electrical Engineering (GPA: 4.19/4.5)

Winnipeg, Canada

Awards: Undergraduate Research Award (2025), Dean's Honour List (2023–2026), 3rd Place Undergraduate Research (2025)